

I am writing to express my interest in the PhD position titled “*trusted open hardware and cybersecurity for Software-Defined Vehicles*” at Politecnico di Torino. I have a background and passion in Software Engineering, Machine Learning, and Autonomous Agents, and I believe I can contribute to your research group with my skills and enthusiasm.

I hold an M.Sc. degree in IT Engineering from the University of Isfahan, where I graduated with a GPA of 17.99/20 and ranked 2nd among my graduating class. I also hold a B.Sc. degree in Computer Software Engineering from Lorestan University, where I graduated as the top student in my graduating class with a GPA of 17.02/20. During my studies, I have gained proficiency in various programming languages, such as Java and Python, and I have completed diverse projects and courses related to Multi-Agent Systems, Autonomous Agents, Data Mining, and Machine Learning.

For my master’s thesis, I focused on a topic related to Multi-Agent Systems: “*Effectively Modeling Single-Peaked Opponents in Automated Bilateral Negotiations*”. This research aimed to evaluate the capabilities of various ANAC agents [1] in opponent modeling, utilizing Machine Learning techniques to estimate the opponent’s behavior during automated negotiations. This work involved developing cooperative decision-making models to improve negotiation outcomes, aligning closely with the principles of cooperative decision-making, where agents work together to find mutually beneficial solutions. I authored three papers [2-4] based on this thesis, which are also listed in my CV. Through this research, I gained valuable experience in designing, implementing, and evaluating automated negotiation agents.

After completing my post-graduation, I worked as a Research Assistant at the University of Isfahan from December 2022 to October 2024. During this time, I co-authored a paper related to the *EU-Iran STEM/UNITEL* project 2023, which enhanced student engagement and learning outcomes through anonymous feedback systems [5]. I also co-authored the paper “*Combinatorial Auction Based on Social Choice in the Internet of Things*” [6] investigating social choice mechanisms to optimize resource allocation and decision-making processes in IoT environments. In December 2024, I was awarded a research grant from the University of Udine for the project “*Towards Digital and Smart Agriculture: Use of AI and Bioinformatic Approaches to Support the Quality of the Agrifood Chain*”. Since then, I have been working as a researcher on a one-year initiative applying AI and machine learning to agricultural challenges, with a focus on genetic improvement. My work involved the processing and analysis of phenomic data derived from hyperspectral wavelength measurements to develop predictive models for detecting nutrient deficiencies in cultivated plants. I was also engaged in the analysis of microbiome data to model plant–microorganism interactions, as well as in the processing of transcriptomic data from tomato plants to support the characterization of plant responses to iron deficiency. It led to the preparation of two research papers that are now being prepared for the peer-review process. Being part of this multidisciplinary project, I have been gaining valuable experience collaborating with experts from various fields, which is enhancing my ability to approach complex problems from diverse perspectives and find innovative solutions. Beyond my formal education and research experience, I am committed to continuous learning. To further strengthen my expertise in Machine Learning, I have completed specialized coursework and honed my programming proficiency in Python, ensuring that I stay engaged with the latest advancements.

I have a deep appreciation for learning and a strong interest in research. I enjoy writing and have contributed to scientific publications and technical documentation. What attracts me to this PhD opportunity is the group’s focus on computing challenges that arise when hardware, intelligent algorithms, and domain-specific science come together. I am especially interested in resilient systems for safety-critical environments and in advanced computing methods that support discovery in complex biological processes. These themes resonate with my experience in machine learning and explainability, as well as with my current work in bioinformatics, where interdisciplinary collaboration is essential. Although computer architectures have not been the main focus of my previous research, I find their conceptual depth and practical impact compelling. I am eager to contribute to a team that values cross-domain ideas and to grow in an environment where innovation emerges from connecting diverse perspectives. With my background in applied machine learning and my interest in secure and reliable computing, I believe I can bring both creativity and rigor to the group’s work while continuing to develop as a researcher.

Thank you for considering my application. I look forward to the opportunity to discuss how I can contribute to your team.

With respect, Fatemeh Hassanvand

References

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3. Hassanvand, F. and F. Nassiri-Mofakham, *Automated Negotiation Agents in Modeling Gaussian Bidders*. *AUT Journal of Modeling and Simulation*, 2023. **55**(1): p. 3-16.
4. Hassanvand, F., F. Nassiri-Mofakham, and K. Fujita, *Automated Negotiation Agents for Modeling Single-Peaked Bidders: An Experimental Comparison*. *Information*, 2024. **15**(8): p. 508.
5. Nassiri-Mofakham, F., F. Hassanvand, and N. Mojoodi. *Implementing Double-Blind Peer Assessment in Virtual Classroom*. in *2024 11th International and the 17th National Conference on E-Learning and E-Teaching (ICeLeT)*. 2024. IEEE.
6. Esmaeili, M., F. Nassiri-Mofakham, and F. Hassanvand. *Combinatorial Auction Based on Social Choice in the Internet of Things*. in *2023 14th International Conference on Information and Knowledge Technology (IKT)*. 2023. IEEE.